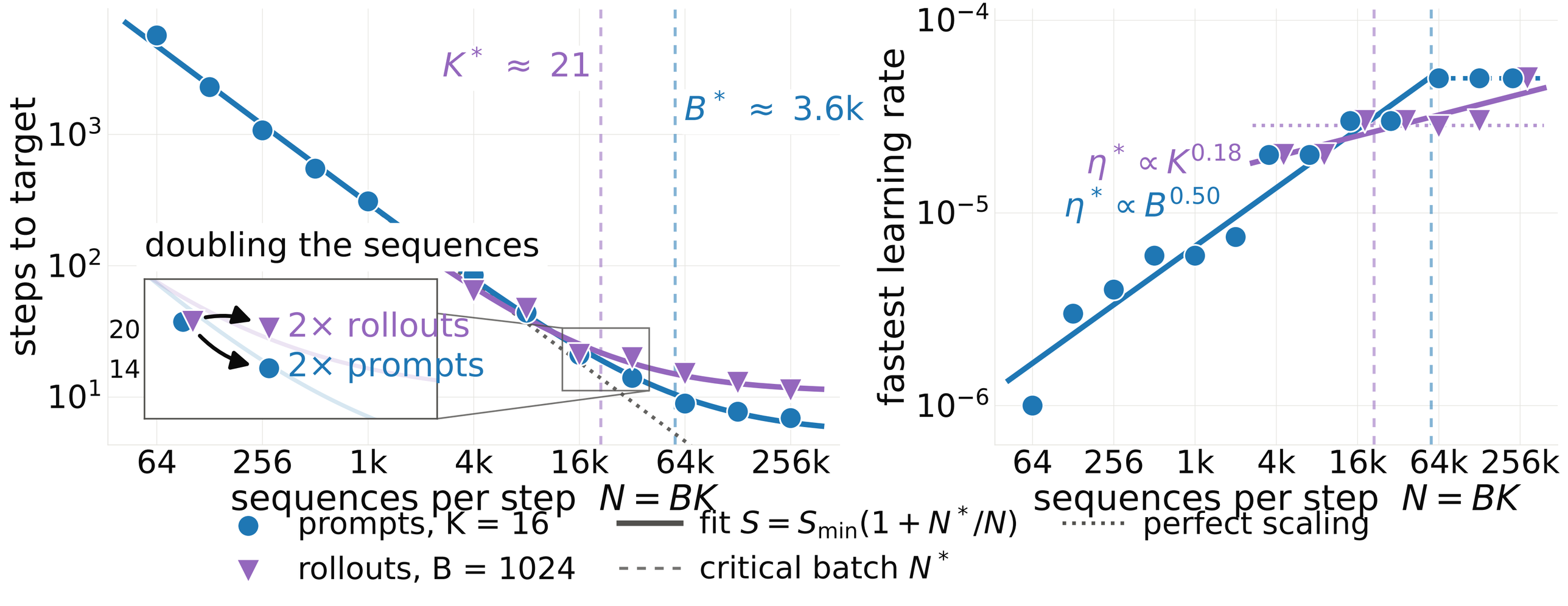
$K^* \approx 21$
$B^* \approx 3.6k$
steps to target
10^3
10^2
10^1
doubling the sequences
20
14
2× rollouts
2× prompts
64
256
1k
4k
16k
64k
256k
sequences per step $N = BK$
fastest learning rate
10^{-4}
10^{-5}
10^{-6}
$\eta^* \propto K^{0.18}$
$\eta^* \propto B^{0.50}$
64
256
1k
4k
16k
64k
256k
sequences per step $N = BK$
prompts, K = 16
rollouts, B = 1024
fit $S = S_{\min}(1 + N^*/N)$
critical batch N^*
perfect scaling